



IF3170 Inteligensi Artifisial

Institut Teknologi Bandung

Intelligent Agent

Semester I TA 2026/2027





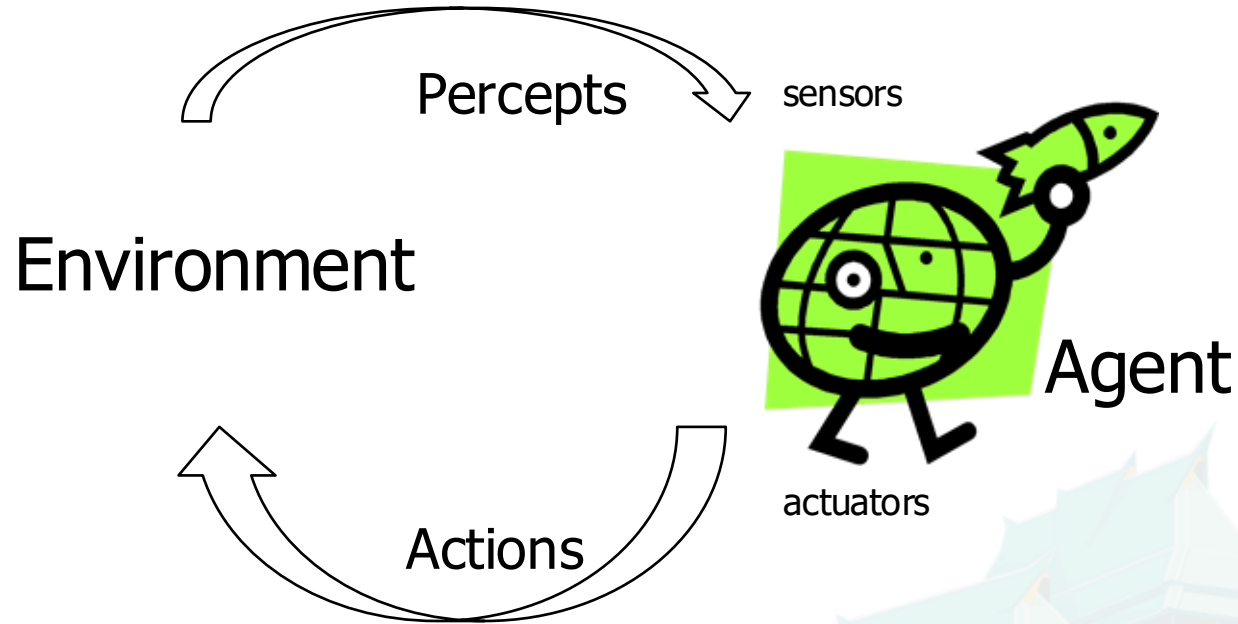
Outline

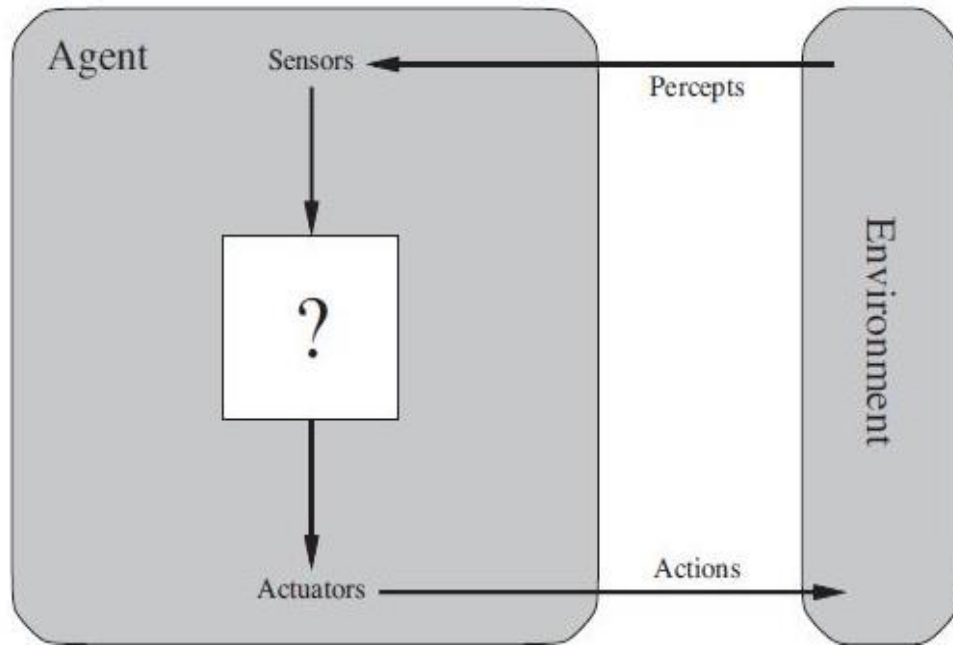
- Review: What is AI (4 Approaches)
- Agents Model
- PEAS
- Task Environment : 6 task environment types
- Agent Types
- Agent Level:
 - Problem Solving Agent
 - Knowledge Based Agent
 - Learning Agent





Agent & Environment







Agents

[1]: Software that **gathers** information about an environment and takes **actions** based on that information.

A robot

A factory

A web shopping program

...

[2]: computer system that is *situated* in some *environment*, and that is capable of *autonomous action* in this environment in order to meet its design objectives

-> **Computational agents that behave autonomously**



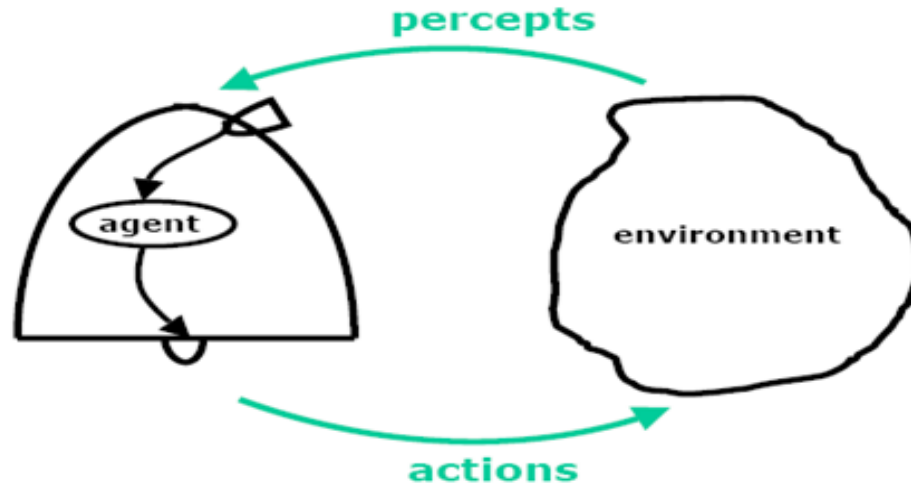
Agents

- Anything that can be viewed as **perceiving** its environment through **sensors** and **acting** upon that environment through **effectors**.
 - A robot
 - A factory
 - A web shopping program
 - ...
- > **Computational agents that behave autonomously**
 - Human agent: eyes, ears, and other organs for sensors; hands, legs, mouth, and other body parts for actuators
 - Robotic agent: cameras and infrared range finders for sensors; various motors for actuators



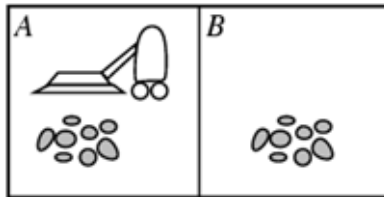
The Agent & the Environment

- How do we begin to formalize the problem of building an agent?
 - Make a dichotomy between the agent and the environment





Example: Vacuum-cleaner world



- Percepts: location and contents, e.g., [A, Dirty]
- Actions: *Left*, *Right*, *Suck*, *NoOp*
- Very simple function: if the current square is dirty, then suck; otherwise, move to the other square



Simple Agent Function

Percept sequence	Action
<i>[A, Clean]</i>	<i>Right</i>
<i>[A, Dirty]</i>	<i>Suck</i>
<i>[B, Clean]</i>	<i>Left</i>
<i>[B, Dirty]</i>	<i>Suck</i>
<i>[A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>
<i>[A, Clean], [A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2. The agent cleans the current square if it is dirty, otherwise it moves to the other square. Note that the table is of unbounded size unless there is a restriction on the length of possible percept sequences.



Simple Agent Program

function REFLEX-VACUUM-AGENT(*[location,status]*) **returns** an action

if *status* = *Dirty* **then return** *Suck*
else if *location* = *A* **then return** *Right*
else if *location* = *B* **then return** *Left*

Figure 2.8 The agent program for a simple reflex agent in the two-location vacuum environment. This program implements the agent function tabulated in Figure 2.3.

Is this “good behaviour”?

Depends on the performance measure : rational agent



Rational Agent

- For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.
- Performance measure: An objective criterion for success of an agent's behavior
- E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.



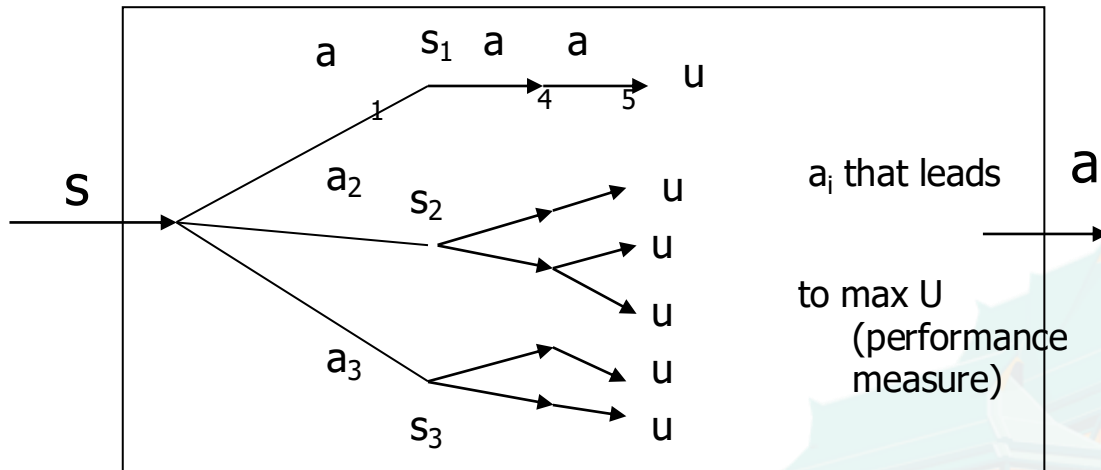
Rationality

- A rational agent takes actions it believes will achieve its goals.
 - Assume I don't like to get wet, so I bring an umbrella. Is that rational?
 - Depends on the weather forecast and whether I've heard it. If I've **heard the forecast** for rain (and I believe it) then **bringing the umbrella is rational**.
- Rationality $\neq$ omniscience
 - Assume the most recent forecast is for rain but I **did not listen** to it and I **did not bring** my umbrella. Is that rational?
 - Yes, since I did not know about the recent forecast!
- Rationality $\neq$ success
 - Suppose the **forecast is for no rain** but I **bring my umbrella** and I use it to defend myself against an attack. Is that rational?
 - No, although successful, it was done for the wrong reason.
- The agent might not be able to compute the best action (subject to its beliefs and goals).
- So, we want to use limited rationality: “**acting in the best way** you can subject to the **computational constraints** that you have”.



Planning Agent Policy

- Planning is explicitly considering future consequences of actions in order to choose the best one.
- So, planning is the process of generating possible sequences of actions, simulating their consequences, picking which is the best and committing to one of these actions.





Limited Rationality (2)

- The (limited rational) agent design problem:
Find $P^* \rightarrow A$
 - mapping of sequences of percepts to actions
 - maximizes the utility of the resulting sequences of states
 - subject to our computational constraints
- To design an agent : specify PEAS (task environment)





Task Environment

- PEAS: Performance measure, Environment, Actuators, Sensors
- Must first specify the setting for intelligent agent design
- Consider, e.g., the task of designing an **automated taxi driver**:
 - Performance measure: Safe, fast, legal, comfortable trip, maximize profits
 - Environment: Roads, other traffic, pedestrians, customers
 - Actuators: Steering wheel, accelerator, brake, signal, horn
 - Sensors: Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard



Another Example of PEAS

Medical diagnosis system Agent

- Performance measure: Healthy patient, minimize costs, lawsuits
- Environment: Patient, hospital, staff
- Actuators: Screen display (questions, tests, diagnoses, treatments, referrals)
- Sensors: Keyboard (entry of symptoms, findings, patient's answers)



Another Example of PEAS

Part-picking robot Agent

- Performance measure: Percentage of parts in correct bins
- Environment: Conveyor belt with parts, bins
- Actuators: Jointed arm and hand
- Sensors: Camera, joint angle sensors





Properties of Task Environment

- **Fully observable** (vs. partially observable): An agent's sensors give it access to the complete state of the environment at each point in time.
- **Deterministic** (vs. stochastic): The next state of the environment is completely determined by the current state and the action executed by the agent.
- **Episodic** (vs. sequential): The agent's experience is divided into atomic "episodes" (each episode consists of the agent perceiving and then performing a single action), and the choice of action in each episode depends only on the episode itself.



Properties of Task Environment (2)

- **Static** (vs. dynamic): The environment is unchanged while an agent is deliberating. (The environment is **semidynamic** if the environment itself does not change with the passage of time but the agent's performance score does)
- **Discrete** (vs. continuous): A limited number of distinct, clearly defined percepts and actions.
- **Single agent** (vs. multiagent): An agent operating by itself in an environment.

This is not strictly a property of the environment

- **Known** (vs Unknown): this distinction refers not to the environment itself but to the agent's (or designer's) state of knowledge about the "laws of physics" of the environment



Example of Task Environment

	Chess with a clock	Chess without a clock	Taxi driving
Fully observable	Yes	Yes	No
Deterministic	Deterministic	Deterministic	No
Episodic	No	No	No
Static	Semi	Yes	No
Discrete	Yes	Yes	No
Single agent	No	No	No

- It determines the agent design
- The real world is (of course) partially observable, stochastic, sequential, dynamic, continuous, multi-agent



Back to Agent Programs

- An agent is completely specified by the agent function mapping percept sequences to actions
- One agent function (or a small equivalence class) is rational
- Aim: find a way to implement the rational agent function concisely
- We can use table lookup that maps sequence of percepts to action (Figure 2.3 slide 9)





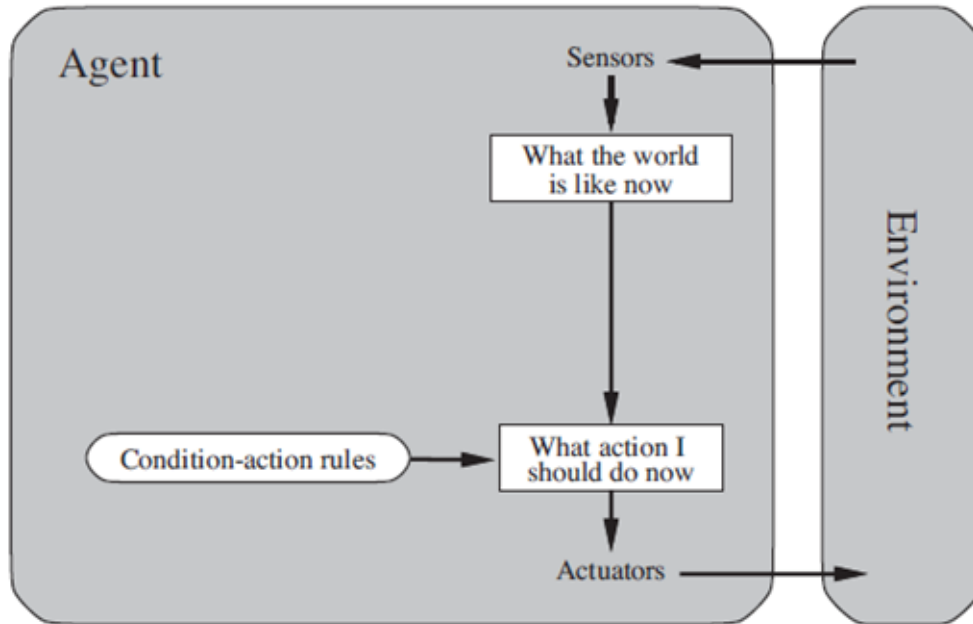
Table-lookup agent

- Drawbacks:
 - Huge table
 - Take a long time to build the table
 - No autonomy
 - Even with learning, need a long time to learn the table entries
- Alternatives : reflex agent: agent respond very flexible to a very broad range of stimuli
 - Drawback: could not store the response
- Next: Basic types of Agent Programs





Simple Reflex Agent



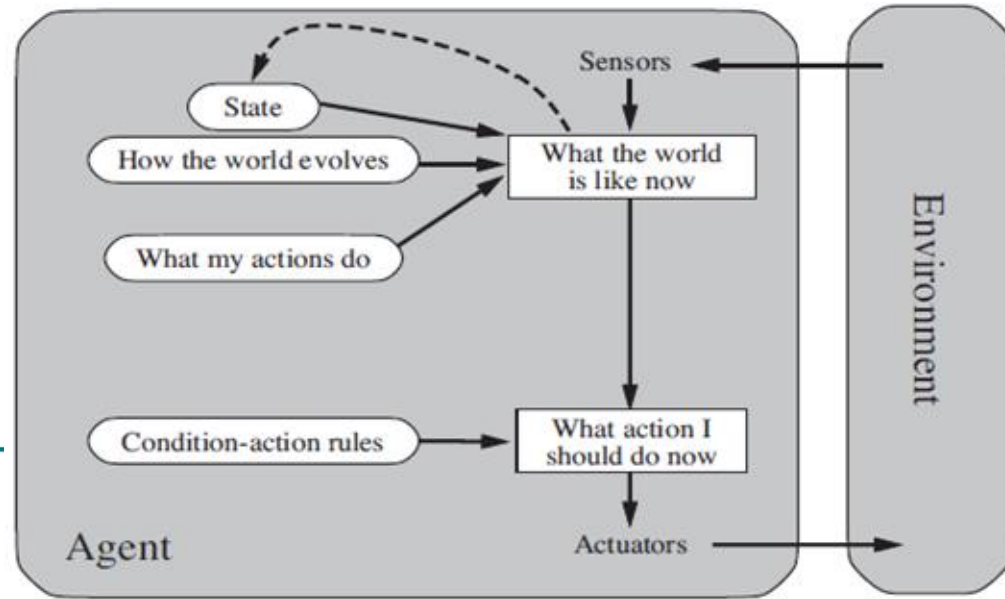
function SIMPLE-REFLEX-AGENT(*percept*) **returns** an action
persistent: *rules*, a set of condition–action rules

```
state ← INTERPRET-INPUT(percept)  
rule ← RULE-MATCH(state, rules)  
action ← rule.ACTION  
return action
```





Model based Reflex Agent



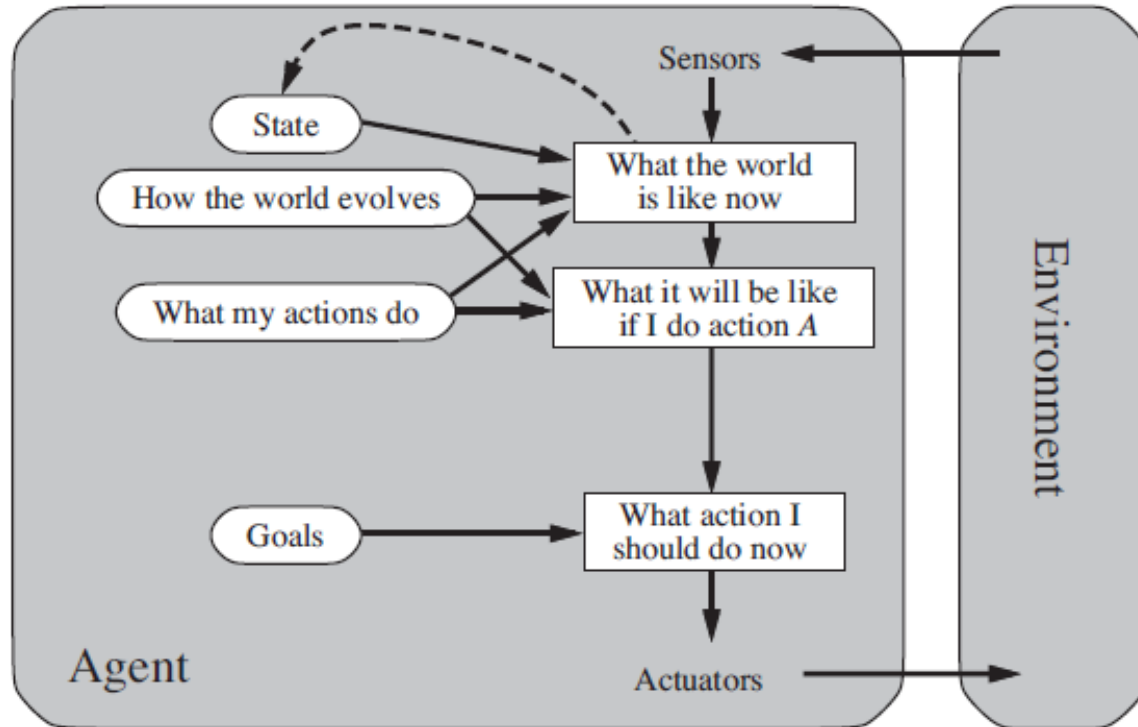
function MODEL-BASED-REFLEX-AGENT(*percept*) **returns** an action
persistent: *state*, the agent's current conception of the world state
transition_model, a description of how the next state depends on
the current state and action
sensor_model, a description of how the current world state is reflected
in the agent's percepts
rules, a set of condition-action rules
action, the most recent action, initially none

```
state ← UPDATE-STATE(state, action, percept, transition_model, sensor_model)
rule ← RULE-MATCH(state, rules)
action ← rule.ACTION
return action
```

Figure 2.12 A model-based reflex agent. It keeps track of the current state of the world, using an internal model. It then chooses an action in the same way as the reflex agent.

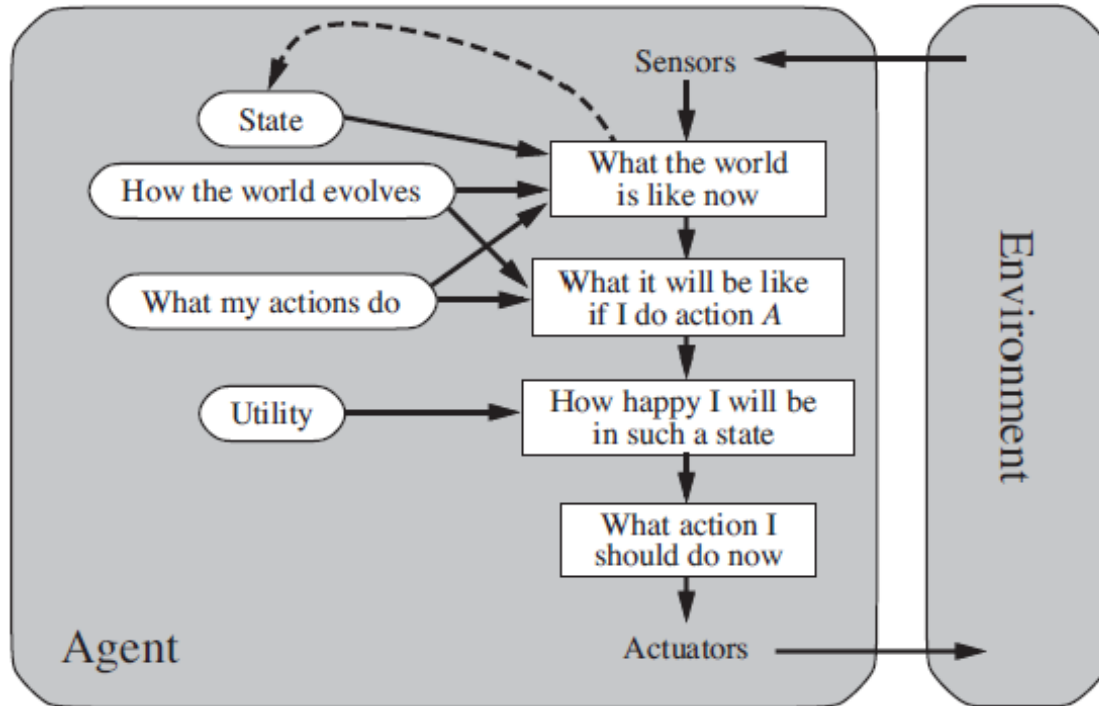


Goal based Agent





Utility based Agent



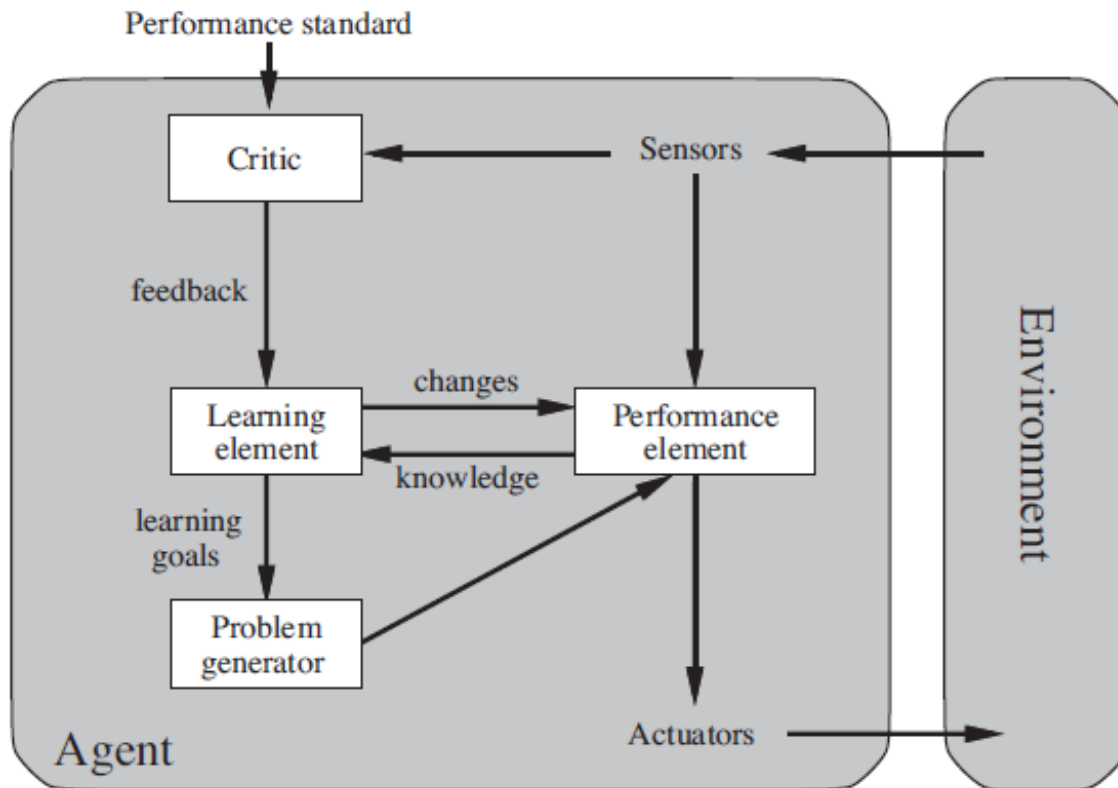


Learning based Agent

Critic tells the learning element how well the agent is doing with respect to a fixed performance standard

The **learning element** (responsible for making improvement) uses feedback from the critic on how the agent is doing and determines how the performance element should be modified to do better in the future

Problem Generator is responsible for suggesting actions that will lead to new and informative experiences



Performance element responsible for element selecting external actions. The performance element is what we have previously considered to be the entire agent: it takes in percepts and decides on actions



AI Agents vs Agentic AI

Source: Ranjan Sapkota, Konstantinos I. Roumeliotis, Manoj Karkee, AI Agents vs. Agentic AI: A Conceptual taxonomy, applications and challenges, Information Fusion, (<https://www.sciencedirect.com/science/article/pii/S1566253525006712>)

Feature	AI Agents	Agentic AI
Definition	Autonomous software programs that perform specific tasks.	Systems of multiple AI agents collaborating to achieve complex goals.
Autonomy Level	High autonomy within specific tasks.	Higher autonomy with the ability to manage multi-step, complex tasks.
Task Complexity	Typically handle single, specific tasks.	Handle complex, multi-step tasks requiring coordination.
Collaboration	Operate independently.	Involve multi-agent collaboration and information sharing.
Learning and Adaptation	Learn and adapt within their specific domain.	Learn and adapt across a wider range of tasks and environments.
Applications	Customer service chatbots, virtual assistants, automated workflows.	Supply chain management, business process optimization, virtual project managers.

Aspect	AI Agent	Agentic AI	Generative Agent
Primary Capability	Task execution	Autonomous goal setting	Content generation
Planning Horizon	Single-step	Multi-step	N/A (content only)
Learning Mechanism	Rule-based or supervised	Reinforcement/meta-learning	Large-scale pre-training
Interaction Style	Reactive	Proactive	Creative
Evaluation Focus	Accuracy, latency	Engagement, adaptability	Coherence, diversity



Leveling of Agent

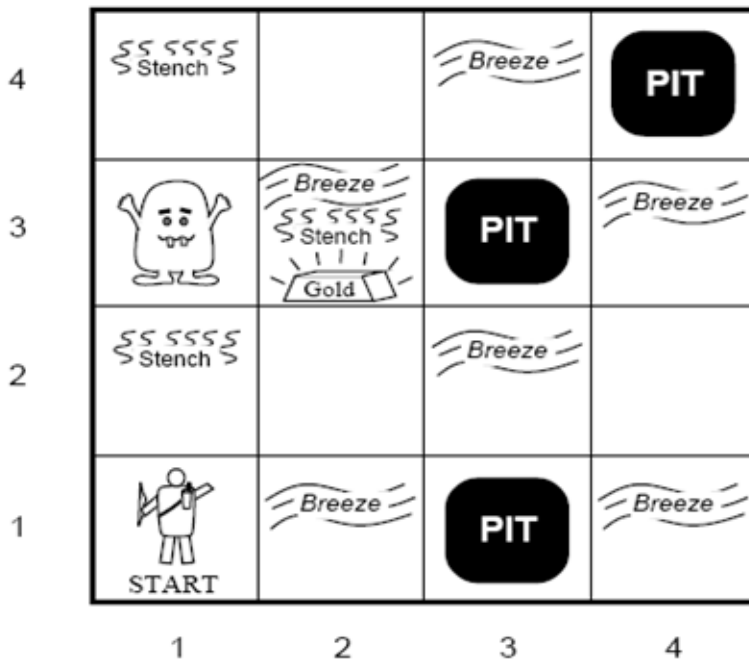
- Problem Solving Agent
- Knowledge Based Agent
- Learning Agent

Let's see the difference from The Wumpus World





The Wumpus World (1)



- Cave with rooms
- Wumpus eats anyone who enters its room
- Wumpus can be shot by an agent, but the agent has only one arrow
- Pit will trap anyone, except for the wumpus
- Agent can find gold heap



The Wumpus World (2)

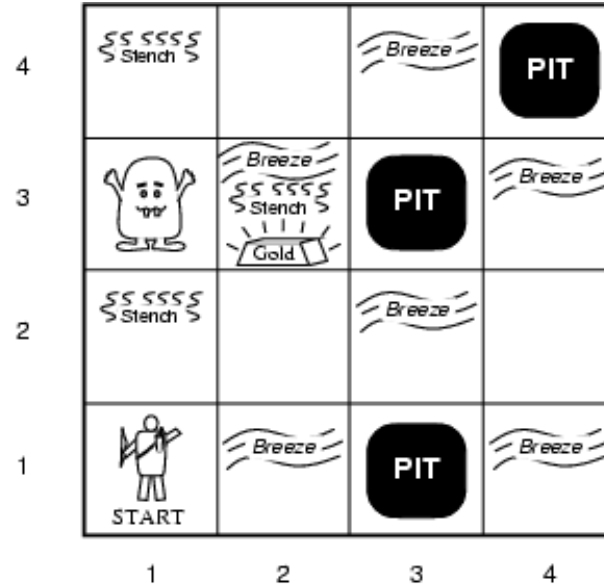
- Sensors: sensor to capture Stench, Breeze, Glitter, Bump, Scream
Actuators: motor to move Left, Right, Forward, hands to Grab, Release, and Shoot arrow
- Performance measure
 - gold +1000, death -1000
 - -1 per step, -10 for using the arrow





Problem Solving Agent (in Wumpus World)

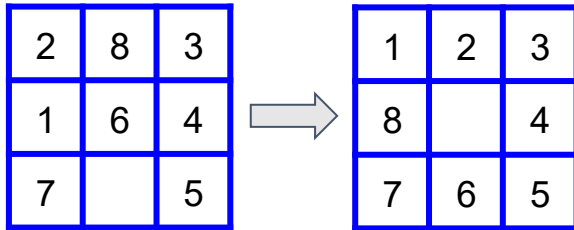
- Agent has information about all of the states in Wumpus World, e.g:
 - Wumpus in room (1,3)
 - There are pits in room (3,1), (3,3), (4,4)
 - The gold is in room (2,3) : this is the goal
 - Etc.
- Initial state: agent in room (1,1)
- Agent has to 'search', the path that can lead agent to the goal, as fast as possible
- There are many searching algorithms, and each algorithm is suitable for certain problem:
 - DFS, BFS, IDS, UCS
 - A*, Greedy Best First,
 - Minmax search,
 - Genetic Algorithm, Hill Climbing, Simulated Annealing,
 - Etc...



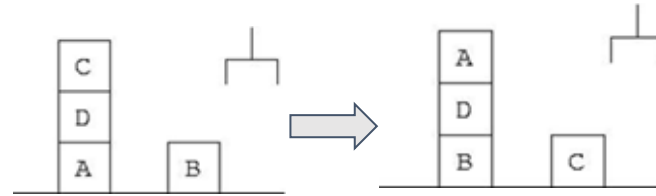


Other Example Problem Solving Agent: search from initial to goal state

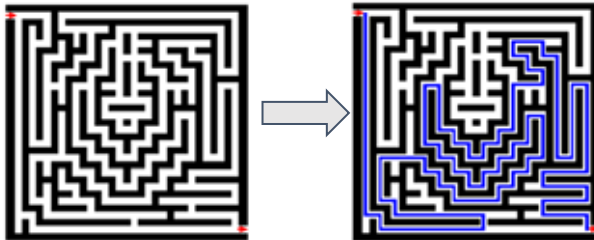
N puzzle problem



Block-world problem



Maze problem



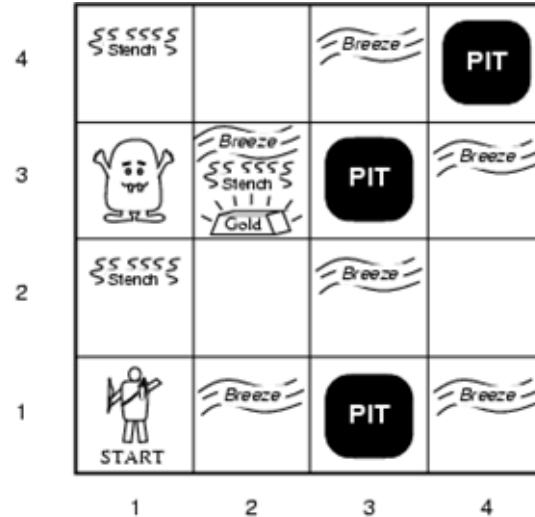
Path-finding problem





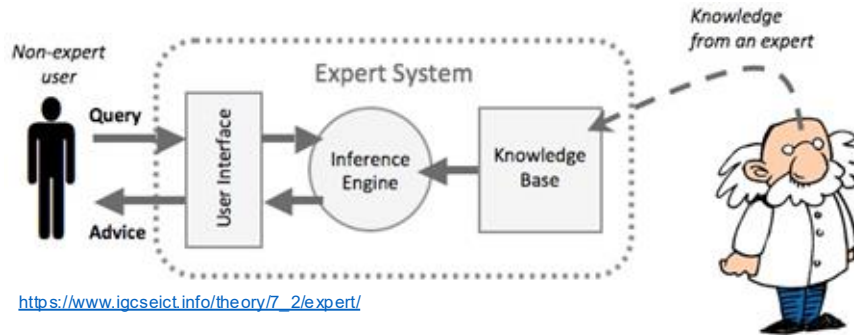
Knowledge Based Agent (in Wumpus World)

- Agent doesn't have information about all of the states in Wumpus World. It only has 'basic knowledge/ premises':
 - Squares adjacent to wumpus are smelly
 - Squares adjacent to pit are breezy
 - Glitter iff gold is in the same square
 - Shooting kills wumpus if you are facing it
 - Shooting uses up the only arrow
 - Grabbing picks up gold if in same square
 - Releasing drops the gold in same
- When agent perceives a state in a room, it will try to reason new facts/ states, this is how agent will step by step collecting all of the states of wumpus world in order to achieve its goal
- Reasoning has to be done by Agent: by deducting the premises with perceived fact.





Other Example Knowledge-based agent

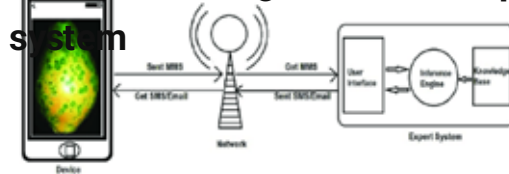


https://www.icgcseict.info/theory/7_2/expert/



<http://www.classgist.com/blogs/169/online-expert-system-for-car-repair-and-maintenance-php-source-codes.aspx>

vision-based agro-medical expert system



https://www.researchgate.net/figure/The-architecture-of-online-machine-vision-based-agro-medical-expert-system_fig2_325849466

TB disease detection application

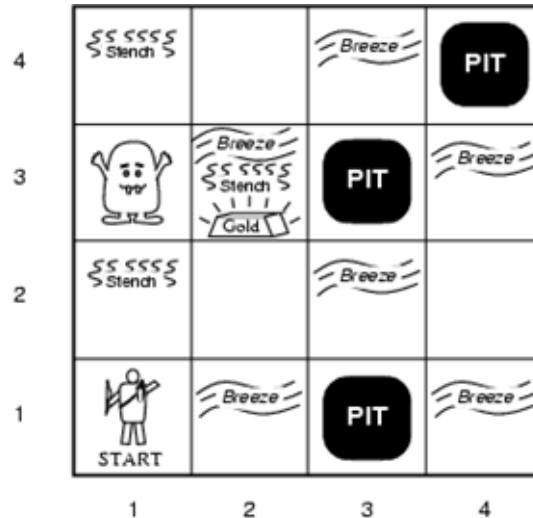


<https://www.sciencedirect.com/science/article/pii/S0957417418304214>



Learning Agent (in Wumpus World)

- Agent doesn't have the information of all of the states and doesn't even have the basic knowledge of the wumpus world.
- Agent plays several times (perhaps dies several times)
- The **observation data** from playing several times is the 'input' for learning process
- The result of the learning process, agent will have basic knowledge, e.g. Squares adjacent to pit are breezy
- There are many learning algorithms, that suitable for certain purposes, and the 'availability' of the data/feedback
 - Supervised learning
 - Unsupervised learning
 - Reinforcement learning





Other Example Learning Agent

Hand-writing recognition



<https://www.imore.com/i-want-handwriting-recognition-iphone-and-ipad-even-if-it-stinks>

Speech

Debat Capres #5
Ekonomi dan Kesejahteraan Sosial, Keuangan dan Investasi,
serta Perdagangan dan Industri
13 April 2019 | Status Pengolahan Data | Selesai

Transkrip Lengkap

No.	Pembicara	Waktu (WB)	Transkrip
1	Balques Marisang	20:04	baik kami meminta hadirin baik kami meminta hadirin berdiri karena kita akan bersama-sama menyanyikan lagu kebangsaan Indonesia raya
2	pembicara	20:04	(Lagu Kebangsaan Indonesia Raya)
3	Balques Marisang	20:06	pak Maruf Amin untuk berisalam juga dengan Prabowo Subianto dan bapak Salahudin iya silakan baik tidak sabar rasanya Pak silakan Pak monggo ditempati tempatnya satu-satu
4	Balques	20:07	baik sebelum memulai acara karena sudah siap semua tapi ada yang juga sangat penting untuk memulai acara yang besar ini kita akan membacakan doa menundukan sejenak kepala kita dan yang akan memimpin doa adalah Profesor Kiki Hui Nanasukhin Umar M.S. Ph.D. Insan Rasis Masjid Istiqlal

<http://debatcapres.bahasakita.co.id/>

Face recognition



https://www.nec.com/en/global/solutions/safey/face_recognition/NeoFaceWatch.html



References

- Stuart J Russell & Peter Norvig, Artificial Intelligence: A Modern Approach, 4th Edition, Prentice-Hall International, Inc, 2022, Textbook
Site: <http://aima.cs.berkeley.edu/> (4th Edition)





Thank You

